

Testing

1. Strategy

LearnHub is tested at several levels, from fast checks of single classes up to the published application running in a real browser. Every level runs automatically in GitHub Actions on every push that touches the application.

Level	What it proves	Tools	Where
Unit	Grading, progress, validation rules, content safety, file signatures and names	xUnit v3	<code>tests/LearnHub.Tests/Unit</code>
Data model	The EF Core model creates a valid SQLite schema, a create script that enforces the integrity rules, and only the delete rules the design intends	xUnit, EF Core	<code>tests/LearnHub.Tests/Data</code>
Service	Business rules against a real (in-memory SQLite) database	xUnit, EF Core	<code>tests/LearnHub.Tests/Services</code>
Integration	The whole application in memory: routing, authentication, authorisation, antiforgery, forms, views, database	<code>WebApplicationFactory</code>	<code>tests/LearnHub.Tests/Integration</code>
Link crawl	Every link reachable by a guest, a student and an administrator returns a valid page	<code>WebApplicationFactory</code>	<code>NavigationCrawlTests</code>
Container	The Docker image Railway builds, started the way Railway starts it — Production settings, <code>PORT=8080</code> , a volume at <code>/data</code> : migrations apply once, demo data is seeded, the administrator is created, the smoke test passes, and after a restart migrations are not re-applied, which proves the volume persists the SQLite file	Docker, <code>scripts/smoke-test.sh</code>	<code>ci.yml</code> job <code>container</code>

Level	What it proves	Tools	Where
Browser	Guest, student and admin journeys on desktop, tablet and phone sizes; console errors, failed requests, broken images, sideways scrolling	Playwright	<code>tests/e2e/specs</code>
Accessibility	Automated WCAG 2.2 A and AA rules on public, student and admin pages	axe-core with Playwright	<code>tests/e2e/specs/accessibility.spec.js</code>
Smoke	A running instance (the container in CI, and the deployed application once Railway is deployed) serves pages, links, protected areas and security headers	<code>scripts/smoke-test.sh</code>	<code>ci.yml</code> job <code>container</code>
Manual	Things automation does not judge well: readability, keyboard flow, visual quality	Checklist in section 7	This document

`tests/LearnHub.Tests/Infrastructure` holds no tests of its own: it is the shared test host — the in-memory application factory, the isolated SQLite database helper, the test environment and the browser client extensions.

Test data is isolated: each integration test class gets its own in-memory SQLite database with migrations and demo data applied, so tests can run in parallel and never touch a developer's database. Test accounts use synthetic passwords defined in the test factory; CI generates random passwords for the browser tests.

The three CI jobs are `build-and-test` (the whole .NET suite), `container` (the Docker image started the way Railway starts it, plus the smoke test) and `e2e` (Playwright against a Production instance over HTTPS).

2. How to run the tests

See [README – Running the tests](#). In short:

```
dotnet test --solution LearnHub.sln
```

3. Latest results

Evidence: measured on 14 September 2026 on the working branch `feature/railway-firebase-sqlite` with `dotnet test LearnHub.sln -c Release`. Every job uploads its evidence as an artifact — the TRX result files, the Playwright report and screenshots, and the container log — so the figures below can be reproduced from the [Actions page](#) or locally.

Job	Result
build-and-test	Release build with warnings treated as errors: 0 warnings, 0 errors; 211 of 211 tests passed , 0 failed, 0 skipped (116 methods; theories run once per data row)
container	The image built and started with Production settings, <code>PORT=8080</code> and a volume at <code>/data</code> ; the health check passed; <code>scripts/smoke-test.sh</code> passed; the start-up log showed the migrations applied, the demo data seeded and the administrator account created, with no <code>fail:</code> or <code>crit:</code> lines; after a restart the log showed no migration applied again, so the SQLite file and the Data Protection key ring really live on the volume
e2e	42 of 42 browser checks passed (14 Playwright tests × 3 viewports: desktop 1440×900, tablet 768×1024, mobile Pixel 7), including 9 axe-core accessibility scans covering 26 page views per viewport, with no WCAG A or AA violations; no errors in the application log

4. Automated test inventory

116 test methods produce 211 test cases, because theories run once per data row (Unit 36 methods, Services 23, Data 3, Integration 54).

Class	Methods	Covers
Unit/LogicTests (grader, display format, progress, view model validation)	16	Scoring and the pass mark (rounding down, unanswered questions, option ids from other questions, questions weighted by their points), progress width classes and completion derived from counts, initials, human-readable durations and file sizes, category line colours, registration rules (strong matching passwords, accepted terms), question rules (two options, one correct, no duplicates), category icon allow-list
Unit/ContentSafetyTests (video parser, lesson renderer, search pattern)	10	YouTube and Vimeo URLs accepted only in known formats and rebuilt as privacy-enhanced embeds, HTML in lesson text is encoded, only allow-listed formatting is produced, code blocks stay encoded, attacks through formatting markers stay text, LIKE wildcards escaped, long search terms truncated
Unit/FileStorageServiceTests	7	Valid image saved with a random name and detected type, mismatched content rejected, oversize files rejected even with a small declared length, PDFs accepted for documents, paths outside the storage root never resolved, only uploaded thumbnails can be deleted
Unit/DevelopmentSeedPasswordsTests	3	Development generates strong demo passwords once and reuses them from the git-ignored credentials file; configured passwords are never replaced; other environments never generate passwords
Data/DatabaseModelTests	3	SQLite schema creation from the model, the create script enforcing the integrity rules, intentional delete behaviour of key relationships (Restrict and Cascade)

Class	Methods	Covers
Services/CatalogueServiceTests	12	Duplicate category names (case-insensitive), category deletion rules, search and filters return published courses only (including matches in lesson titles and descriptions), unpublished details hidden except for admins, double enrolment and unpublished enrolment rejected, stored progress recalculated when content changes, draft lessons hidden and excluded from progress, leaving a course keeps completions
Services/QuizAndCourseManagementTests	11	Graded attempts store every answer, other students' results are hidden, non-enrolled students cannot take quizzes, quiz pages never contain correct answers, the start time comes from the signed token issued with the quiz page, question points are saved and used for later attempts, foreign option ids are not modified, removing a selected answer keeps attempts, course deletion order
Integration/PublicSiteTests	19	Public pages load, only published courses listed, server-side search over titles, descriptions and lesson titles, category filter and category page, paging beyond the first page, empty search state, course details, friendly 404s, drafts hidden, preview lessons open while others need login, private files protected, security headers, health endpoint, static assets and seeded covers, home map lines, isolated test database
Integration/AuthenticationTests	9	Registration (role, sign-in, password hashing), server-side validation, duplicate email, students and admins sent to their own dashboards after login, generic wrong-password error, local-only return URLs, logout, lockout after five failures, antiforgery rejection
Integration/AuthorizationTests	8	Access matrix for guest, student and admin across protected pages (47 cases over the page lists), 403 access denied page, students cannot post to admin actions even with a valid token
Integration/StudentJourneyTests	5	Register, enrol, study, take the quiz and see the result; tampered answers graded on the server; leaving removes lesson access; profile validation and saving; changing the email address needs the current password
Integration/AdminManagementTests	10	Category create, edit, delete and validation; category with courses protected; course create with cover, edit, publish, delete; resources created, validated, streamed and deleted; quiz publishing needs questions; user roles, deactivation and self-protection; enrolment management; contact inbox; spam honeypot
Integration/NavigationCrawlTests	3	Crawls every internal link as a guest, a student and an administrator: no 500 errors and no broken links

Browser scenarios (14 tests; each runs on desktop 1440×900, tablet 768×1024 and Pixel 7, so 42 checks)

Spec	Scenario
<code>public.spec.js</code>	Home page presents the catalogue (and the map draws one line per category)
<code>public.spec.js</code>	Navigation menu works on every screen size
<code>public.spec.js</code>	Server-side search and category filter
<code>public.spec.js</code>	Course details and a free preview lesson
<code>public.spec.js</code>	Contact form validates in the browser before submitting
<code>public.spec.js</code>	Registration form gives live password feedback
<code>public.spec.js</code>	Unknown pages show the friendly 404 page
<code>public.spec.js</code>	About page
<code>student.spec.js</code>	Dashboard, course progress, lesson and quiz
<code>admin.spec.js</code>	Dashboard and management pages
<code>admin.spec.js</code>	Category create and delete through the forms
<code>accessibility.spec.js</code>	Public pages: home, catalogue, course details, preview lesson, About, Contact, Privacy, login, register, 404
<code>accessibility.spec.js</code>	Student pages: dashboard, My Courses, lesson, quiz history, profile, change password
<code>accessibility.spec.js</code>	Administration pages: dashboard, courses, new course, categories, resources, quizzes, quiz results, enrolments, users, messages

Every browser scenario fails on console errors, failed requests to LearnHub, broken images or horizontal scrolling, and saves full-page screenshots. The app runs from the published build in Production mode over HTTPS, so secure cookies, CSP and error pages behave as in production.

5. Representative test cases

ID	Test case	Steps	Expected	Actual	Result
TC-01	Register with valid data	Fill in the form with a strong password, accept terms, submit	Account created with the Student role, signed in, dashboard shown, password stored as a hash	As expected	Pass
TC-02	Register with a weak password	Enter "short"	Message "Use at least 8 characters with an upper-case letter, a lower-case letter and a number."	As expected; the hint is hidden while the same rule is shown as the error	Pass
TC-03	Duplicate email	Register twice with the same email	One clear error, no second account	As expected	Pass

ID	Test case	Steps	Expected	Actual	Result
TC-04	Lockout	Five wrong passwords	Account locked for 15 minutes with the lockout message	As expected	Pass
TC-05	Open redirect	Log in with <code>returnUrl=https://example.org</code>	Redirect to the user's own dashboard	As expected	Pass
TC-06	Search	Search "sql"	Published courses that match the title, the description or a lesson title are listed with the number found, and drafts never appear	As expected	Pass
TC-07	Preview lesson	Guest opens a free preview, then a locked lesson	Preview shown; locked lesson redirects to login	As expected	Pass
TC-08	Enrol twice	Post the enrol form twice	Second attempt shows "You are already enrolled in this course."	As expected	Pass
TC-09	Quiz tampering	Submit an option id from another question	Recorded as unanswered and counted as incorrect	As expected	Pass
TC-10	Result privacy	Student B opens student A's result URL	404	As expected	Pass
TC-11	Admin access	Student opens <code>/Admin/Courses</code> and posts to an admin action	Access denied (403) and the POST is refused	As expected	Pass
TC-12	Missing antiforgery token	POST to <code>/Account/Login</code> without the token	400 Bad Request	As expected	Pass
TC-13	Disguised upload	Upload an HTML file containing a script, renamed <code>cheat-sheet.pdf</code>	Rejected with "The file content does not match an allowed file type."; nothing stored	As expected	Pass
TC-14	Category with courses	Delete a category that has courses	Deletion refused with an explanation	As expected	Pass
TC-15	Publish empty quiz	Publish a quiz without questions	"Add at least one question before publishing this quiz."	As expected	Pass
TC-16	Last administrator	The only admin tries to deactivate or demote themselves	Refused with a clear message	As expected	Pass
TC-17	Course deletion	Delete a course with attempts	Answers, attempts, quizzes, lessons and enrolments removed; no database error	As expected; the deletion order is deliberate (see ERD)	Pass

ID	Test case	Steps	Expected	Actual	Result
TC-18	Broken links	Crawl all pages as each role	No 404 or 500 responses from internal links	As expected	Pass
TC-19	Responsive layout	Browser tests at 1440, 768 and Pixel 7 widths	No horizontal scrolling, menus usable	As expected (after the fixes in section 6)	Pass
TC-20	Accessibility	axe-core WCAG 2.2 A/AA scan on 26 page views per viewport	No violations	As expected (after the fixes in section 6)	Pass

6. Defects found and fixed

#	Found by	Symptom	Cause	Fix	Commit
1	CI build	xUnit v3 tests would not run on the .NET 10 SDK	VSTest mode no longer supported for this setup	Opted into Microsoft Testing Platform in <code>global.json</code>	<code>32625d6</code>
2	CI build	Razor compile errors in the pagination partial	A variable named <code>page</code> clashed with the reserved <code>@page</code> directive	Renamed the variables	<code>fc83d0a</code>
3	Link crawl test	<code>/Admin/Courses/Details/{id}</code> returned 500 on SQLite	The query used SQL <code>APPLY</code> , which SQLite cannot translate	Rewrote it as separate flat queries	<code>945aed8</code>
4	Admin CRUD test	Deleting a course failed with a "relationship severed" error	EF Core tried to delete tracked dependants with Restrict rules	Delete dependent quiz answers first with <code>ExecuteDelete</code> in a transaction	<code>945aed8</code>

#	Found by	Symptom	Cause	Fix	Commit
5	CI smoke test	Form pages returned 500 in Production over plain HTTP	TLS is terminated by the platform proxy, so the application saw an insecure request and refused its own Secure-only antiforgery cookie	Added <code>UsePlatformProxyHeaders()</code> and made the smoke test send <code>X-Forwarded-Proto</code>	3851e07
6	Browser test (tablet)	Admin pages scrolled sideways by 119 px	Visually hidden table header text escaped the scroll container	Made <code>.table-wrap</code> the containing block	acee776
7	Screenshot review	Admin sidebar invisible on desktop	Bootstrap forces <code>.offcanvas-lg</code> to a transparent background	Restored the dark rail with a more specific rule	acee776
8	Screenshot review	Admin filters always stacked	Invalid CSS grid template (<code>auto-fit</code> combined with <code>auto</code>)	Flexible wrapping row	acee776
9	Browser test (phone)	Lesson pages scrolled sideways by 198 px	Grid column sized to the longest code line	<code>minmax(0, 1fr)</code> column	3851e07
10	Browser test (phone)	About page scrolled sideways	Bootstrap <code>g-5</code> gutters wider than the container padding	<code>g-4 g-lg-5</code>	3851e07
11	Screenshot review	Home map labels overlapped lines and the hub	Label positions did not account for long names	Two-line labels placed clear of the lines; long names shortened	ab9ad5d

#	Found by	Symptom	Cause	Fix	Commit
12	Screenshot review	Home map invisible when animations are disabled	The resting style was hidden and only the animation fill showed it	Keyframes now animate from hidden to a visible resting style	ab9ad5d
13	Accessibility scan	Code blocks and the About table could not be scrolled with the keyboard	Scrollable regions without focusable content	Focusable regions with labels	5de245b
14	Accessibility scan	Two admin top bar links had no name on phones	Their text is hidden below the breakpoint	<code>aria-label</code> on both links	5de245b
15	Build log review	91 build warnings (xUnit1051) in test code	Async calls did not pass the test cancellation token	Token passed everywhere; CI now treats warnings as errors	273ccc6

7. Manual testing checklist

Recorded on 14 September 2026 against the container build of the `feature/railway-firebase-sqlite` working tree, using the Playwright screenshots of every scenario on desktop, tablet and phone plus the automated reports. Items that need a person with a live deployment are marked **To do**; the team completes them after the first Railway deploy.

Area	Check	Result	Evidence or note
Build	Solution builds in Release with no errors and no warnings	Pass	CI build step with <code>-p:TreatWarningsAsErrors=true</code>
Pages	Every public, student and admin page loads without server errors	Pass	Link crawl tests as all three roles
Roles	Guest, student and admin see only their own navigation and pages	Pass	Authorisation tests; screenshots of each role
CRUD	Create, read, update and delete work for courses, categories, resources, quizzes, questions, enrolments, users and messages	Pass	Admin management integration tests; admin browser test
Validation	Client-side messages appear before submit; server repeats every rule	Pass	Contact and registration browser tests; server validation tests
Responsive	Layout correct at 1440, 768 and 412 px; no sideways scrolling	Pass	Browser tests and screenshot review (defects 6–10 fixed)
Console	No JavaScript errors or failed requests	Pass	Browser tests fail on any console error

Area	Check	Result	Evidence or note
Images	No broken images, covers and diagrams load	Pass	Browser tests check every image
Links	No dead internal links	Pass	Link crawl tests
Accessibility (automated)	No WCAG 2.2 A/AA violations found by axe-core	Pass	Accessibility spec in 3 viewports
Visual quality	Consistent design system, readable typography, balanced layouts	Pass	Screenshot review; issues 7, 8 and 11 fixed
Error pages	404 and access denied pages are friendly and branded	Pass	Browser test and integration tests
Security headers	CSP and other headers present	Pass	Integration test and smoke test
Keyboard	Complete journeys (register, enrol, lesson, quiz, admin form) with keyboard only; focus always visible	To do	Focus styles exist for all interactive elements; needs a person to run through
Screen reader	Headings, landmarks, form labels and quiz questions announced correctly (NVDA or VoiceOver)	To do	Automated checks passed; needs a manual pass
Real devices	iPhone and Android phone in portrait and landscape	To do	Emulated Pixel 7 passed
Production	Health, HTTPS redirect, admin login, upload and student journey on Railway	To do	The <code>container</code> job already performs these checks against the image; run the verification table in Deployment after the first Railway deploy